MOVIX

RECOMMENDER SYSTEMS

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START DEMO

IMPLICIT APPROACH

movie_id	title	genres	n_ratings	n_watched	wishlist	recent	top10
924	2001: A Space Odyssey (1968)	Adventure|Drama|Sci-Fi	517	1	1	0	0
6502	28 Days Later (2002)	Action|Horror|Sci-Fi	348	0	0	0	0
51662	300 (2007)	Action|Fantasy|War|IMAX	325	0	0	0	0
4370	A.I. Artificial Intelligence (2001)	Adventure|Drama|Sci-Fi	340	0	0	0	0
344	Ace Ventura: Pet Detective (1994)	Comedy	482	0	0	0	0
19	Ace Ventura: When Nature Calls (1995)	Comedy	350	0	0	0	0
1608	Air Force One (1997)	Action|Thriller	330	0	0	0	0
2791	Airplane! (1980)	Comedy	399	0	0	0	0
588	Aladdin (1992)	Adventure|Animation|Children|Comedy|Musical	536	1	0	0	0
1214	Alien (1979)	Horror|Sci-Fi	611	0	0	0	0
1200	Aliens (1986)	Action|Adventure|Horror|Sci-Fi	541	0	0	0	0
3897	Almost Famous (2000)	Drama	384	0	0	0	0
1225	Amadeus (1984)	Drama	393	0	0	0	0
2858	American Beauty (1999)	Drama|Romance	720	0	0	0	0
1208	Apocalypse Now (1979)	Action|Drama|War	470	0	0	0	0
150	Apollo 13 (1995)	Adventure|Drama|IMAX	580	0	0	0	0
1917	Armageddon (1998)	Action|Romance|Sci-Fi|Thriller	455	0	0	0	0
1784	As Good as It Gets (1997)	Comedy|Drama|Romance	445	0	0	0	0
1517	Austin Powers: International Man of Mystery (1997)	Action|Adventure|Comedy	475	0	0	0	0
2683	Austin Powers: The Spy Who Shagged Me (1999)	Action|Adventure|Comedy	478	0	0	0	0
72998	Avatar (2009)	Action|Adventure|Sci-Fi|IMAX	341	2	0	1	0
1270	Back to the Future (1985)	Adventure|Comedy|Sci-Fi	742	0	0	0	0
2012	Back to the Future Part III (1990)	Adventure|Comedy|Sci-Fi|Western	456	0	0	0	0
33794	Batman Begins (2005)	Action|Crime|IMAX	473	4	1	1	1
153	Batman Forever (1995)	Action|Adventure|Comedy|Crime	411	0	0	0	0
1377	Batman Returns (1992)	Action|Crime	333	4	1	1	1
4995	Beautiful Mind, A (2001)	Drama|Romance	506	0	0	0	0
595	Beauty and the Beast (1991)	Animation|Children|Fantasy|Musical|Romance|IMAX	431	0	0	0	0
2174	Beetlejuice (1988)	Comedy|Fantasy	445	0	0	0	0
2997	Being John Malkovich (1999)	Comedy|Drama|Fantasy	523	0	0	0	0

$$IR = 100 \times \mathrm{top10} + 50 \times \min(\mathrm{n\_watched}, 5) + 30 \times \mathrm{wishlist} + 15 \times \mathrm{recent}$$

LIVE DEMO

Movix

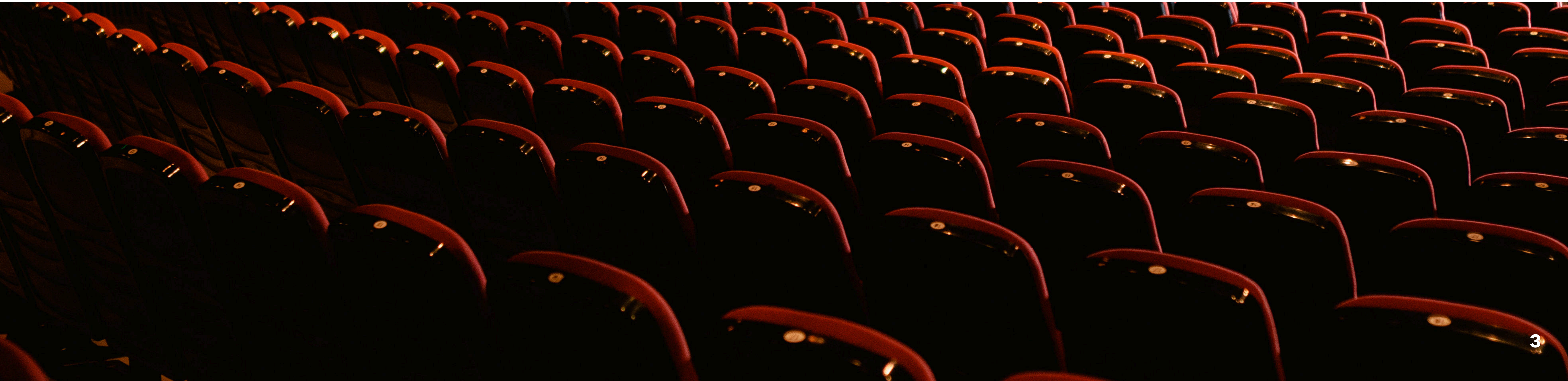
METHODOLOGICAL FRAMEWORK

1. Rating prediction error ➔ 75/25 split

2. Leave-One-Out ➔ full catalogue ranking

3. Negative sampling ➔ 1 positive vs 99 negatives

4. Full-mode ➔ catalogue diversity & novelty (top-40)



RATING ERROR (RMSE / MAE)

Key Takeaways

- **ContentBased Ridge CV →** RMSE: 0.74
  - Direct rating prediction per user
  - L2 regularisation → prevents overfitting
- **Pearson vs Jaccard:** 0.81 vs 0.84
  - Jaccard: ignores rating values
  - Pearson: leverages rating magnitude
- **iALS & BPR →** RMSE: 2.77 & 2.33
  - Not trained to predict explicit ratings
  - RMSE comparison → meaningless here

Model	RMSE ↓	MAE ↓
Content-Based (Ridge CV)	**0.744**	**0.561**
User-Based (Pearson Baseline)	0.805	0.611
User-Based (Tuned, Jaccard)	0.835	0.636
iALS*	2.768	2.577
BPR*	2.328	2.057
BPR+Novelty	3.489	3.334

Table 1: Rating-prediction error on the 75/25 random split.

TOP-N RANKING (LOO FULL-CATALOGUE)

Key Takeaways

- **BPR → HR@10 : 0.137**
  - Only ranking objective → does exactly what we measure
- **Content-Based → last (0.024) despite best RMSE**
  - Accuracy–ranking gap → Predicting well ≠ ranking well
- **Pearson 0.086 vs Jaccard 0.075**
  - Neighbourhood models → middle ground
- **BPR+Novelty → 0.124**
  - Just below BPR

Model	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20
BPR	**0.096**	**0.137**	**0.192**	**0.065**	**0.078**	**0.092**
BPR+Novelty	0.086	0.124	0.167	0.057	0.069	0.080
User-Based (Pearson Baseline)	0.055	0.086	0.129	0.038	0.048	0.059
User-Based (Tuned, Jaccard)	0.050	0.075	0.113	0.032	0.040	0.050
iALS	0.016	0.028	0.051	0.010	0.013	0.019
Content-Based (Ridge CV)	0.015	0.024	0.041	0.010	0.012	0.017

TABLE 2 – Full-catalogue Leave-One-Out ranking. The higher the value, the better.

TOP-N RANKING (NEGATIVE SAMPLING)

Key Takeaways

- **100 items only : 1 positive + 99 random**
  - Very different from Table 2 → that's the point
- **iALS → 1st (HR@10 : 0.857) vs near-last in Table 2**
  - Concentrates scores on few items → easy with 99 randoms
  - But drowns among 9,000 candidates
- **BPR → 2nd (HR@10 : 0.782)**
  - Slight NDCG edge over iALS → ranks target higher when found

Model	HR@5 (*ns*)	HR@10 (*ns*)	HR@20 (*ns*)	NDCG@5 (*ns*)	NDCG@10 (*ns*)	NDCG@20 (*ns*)
iALS	**0.712**	**0.857**	**0.904**	0.485	0.532	0.544
BPR	0.661	0.782	0.866	**0.495**	**0.535**	**0.556**
Content-Based (Ridge CV)	0.226	0.315	0.462	0.158	0.187	0.224
User-Based (Pearson)	0.115	0.222	0.430	0.062	0.097	0.149
User-Based (Jaccard)	0.096	0.200	0.380	0.058	0.091	0.136
BPR+Novelty	0.625	0.751	0.817	0.469	0.511	0.528

TABLE 3 – Negative-sampling Leave-One-Out (He et al., 2017) : the held-out item is ranked against 99 randomly drawn unrated items, a fixed pool of 100 candidates per user.

CATALOGUE DIVERSITY

Key Takeaways

- **Shifting the focus**
- **3 metrics : Coverage / ILD / MIUF**
- **BPR+Novelty → best coverage (51.17%) & highest MIUF (3.627)**
  - Popularity penalty works
  - ILD unchanged vs BPR (0.650 vs 0.655) → Doesn't add variety
- **User-Based → opposite pattern**
  - Highest ILD (≈0.73) → lists internally diverse
  - Coverage collapses (5–6%) & lowest MIUF → same popular films to everyone

Model	Coverage (%) ↑	ILD ↑	MIUF ↑
BPR+Novelty	**51.17**	0.650	3.627
BPR	47.54	0.655	3.275
User-Based (Pearson Baseline)	6.07	**0.728**	1.252
User-Based (Tuned, Jaccard)	5.10	0.727	1.188
Content-Based (Ridge CV)	35.90	0.643	**5.878**
iALS	42.19	0.691	3.302

TABLE 4 – Diversity (ILD) and novelty (MIUF) in full mode (models trained on 100% of interactions, top-40 lists). Coverage is the share of the 8,737-item catalogue appearing in at least one list.

SYNTHESIS

Key Takeaways

- **No single model dominates**
  - BPR → best ranker
  - Content-Based → best RMSE + long tail + cold-start
  - BPR+Novelty → widest catalogue coverage
  - User-Based → middle ground but too popular-focused
  - **4 models, 4 carousels → each where it is strongest**
    - Full space Accuracy, Ranking & Diversity

"The benchmark does not select a ***winner****, it justifies why all four models* ***coexist****."*

THANK YOU

FOR YOUR ATTENTION

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